

9.3 Resistivity

Question Paper

Course	CIE A Level Physics
Section	9. Electricity
Topic	9.3 Resistivity
Difficulty	Hard

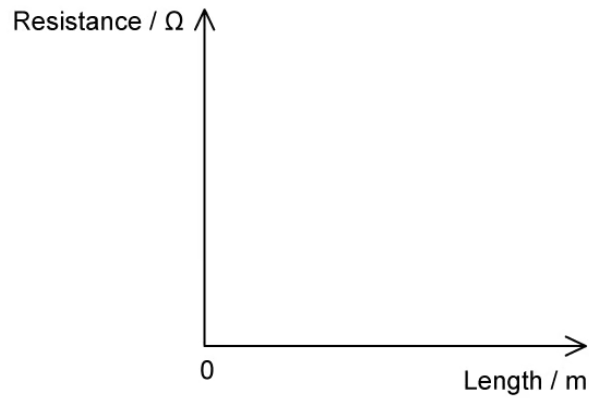
Time allowed: 30

Score: /23

Percentage: /100

Question 1a

A wire of cross-sectional area A is from made metal of resistivity ρ . The wire is extended. Assume that the volume V of the wire remains constant as it extended.

**Fig 1.1**

Sketch on Fig 1.1 how the resistance R of the extending wires varies with the length l .

[2 marks]

Question 1b

Two resistors X and Y have resistance R_X and R_Y respectively. The resistors are connected in series with a battery with electromotive force (e.m.f) E and zero internal resistance, as shown in Fig 1.1.

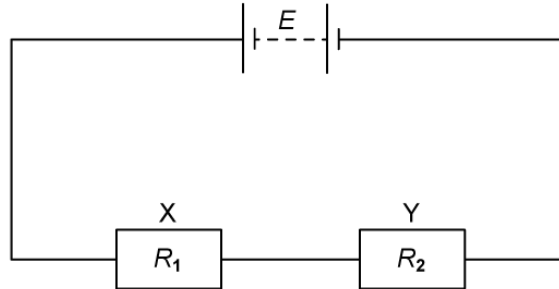


Fig 1.1

The resistors are made from metal wires. Data from the resistors are given in Table 1.1.

Table 1.1

resistor	X	Y
diameter of metal	$\frac{\rho}{3}$	ρ
length of metal	$2l$	l
resistivity of metal	$4d$	d

Use information from Table 1.1 to determine the ratio

$$\frac{\text{power dissipated in Y}}{\text{power dissipated in X}}$$

[3 marks]

Question 1c

Show that the product of the ratio $\frac{\text{power dissipated in Y}}{\text{power dissipated in X}}$ of X and Y connected in series and connected in parallel is 1.

[3 marks]

Question 2a

Thin films of carbon are sometimes used in electronic systems. Typical dimensions of such a film are shown in Fig. 1.1.

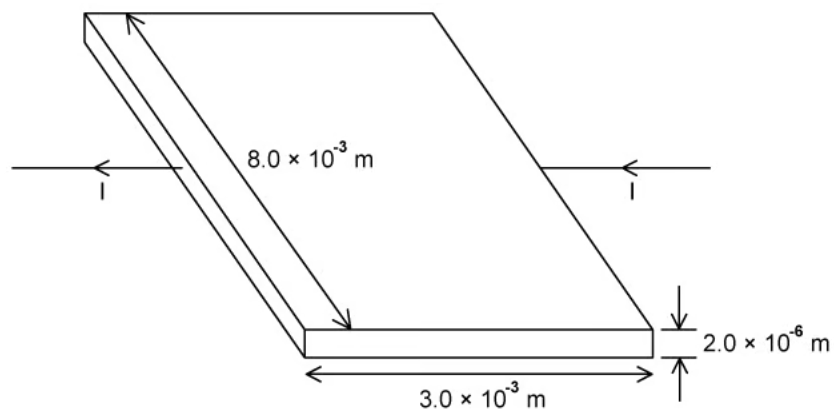


Fig. 1.1.

Calculate the current which passes through the carbon film in Fig. 1.1. for an applied voltage of 2.5 mV.

The resistivity of carbon = $4.0 \times 10^{-5} \Omega \text{ m}$

[5 marks]

Question 2b

The applied voltage is kept constant, but the current is now directed through the carbon film as shown in Fig. 1.2.

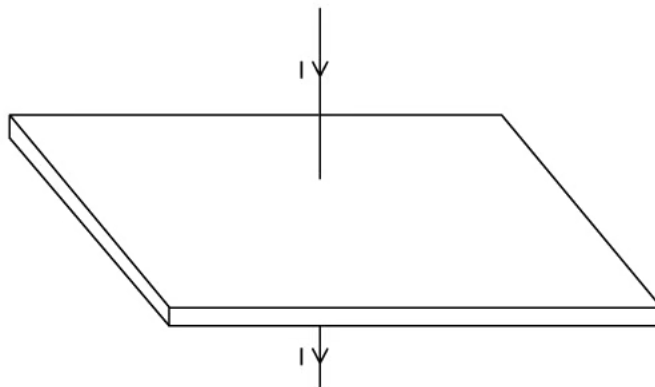


Fig. 1.2.

Show that the current is approximately a million times larger if it is directed through the carbon film as shown in Fig. 1.2.

[6 marks]

Question 2c

A **tensile** force is applied to the carbon film as shown in Fig. 1.2 in a plane that is normal to the current.

Without performing any calculations, discuss how the resistance of the carbon film changes as a result of the applied tensile force.

State any assumptions you make in your answer.

[4 marks]

Model Answers: Hard

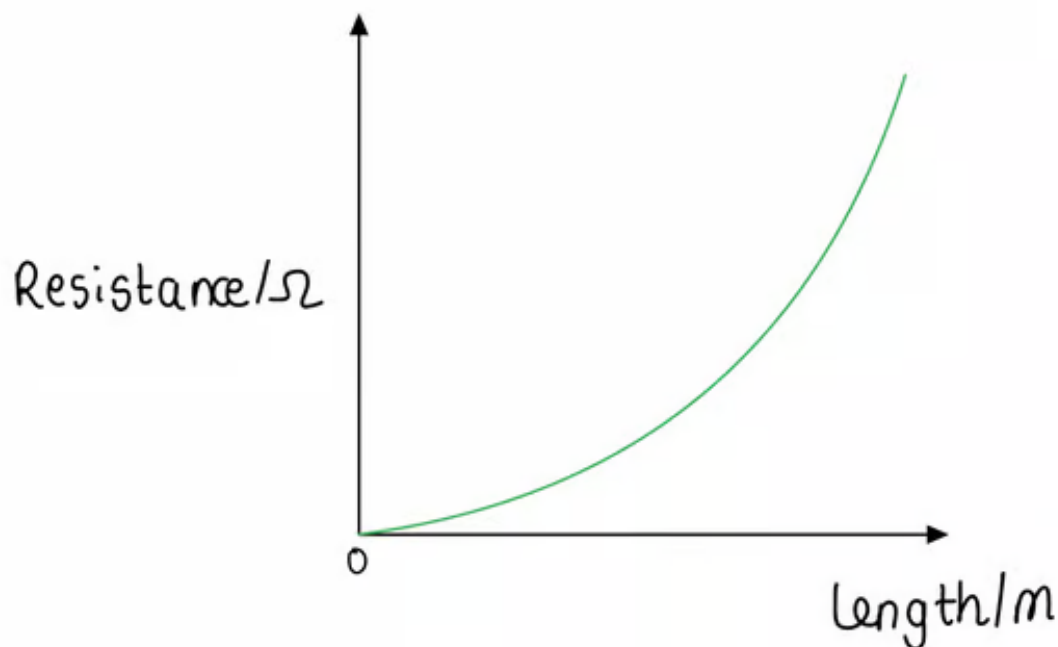
1a

Sketch on Fig 1.1 how the resistance R of the extending wires varies with the length l

Determine the relationship between R and l :

- $R = \frac{\rho l}{A}$
- $V = Al \Rightarrow A = \frac{V}{l}$
- $R = \frac{\rho l}{\frac{V}{l}} = \frac{\rho l^2}{V}$
- Therefore, $R \propto l^2$ [1 mark]

A graph that would score 1 mark should look like:



- Curve lined with increasing gradient; [1 mark]

[Total: 2 marks]

1b

The ratio $\frac{\text{power dissipated in } Y}{\text{power dissipated in } X}$ is:

Determine the equation for the ratio of power dissipated:

- $P = IV = I^2 R$ where $R = \frac{\rho l}{A}$ [1 mark]

$$\bullet \quad \frac{P_Y}{P_X} = \frac{I^2 R_Y}{I^2 R_X} = \frac{\frac{\rho_Y l_Y}{A_Y}}{\frac{\rho_X l_X}{A_X}}$$

Substitute the values from Table 1.1 into the equation and simplify:

$$\bullet \frac{P_Y}{P_X} = \frac{\frac{\frac{\rho l}{\pi d^2}}{4}}{\left(\frac{\rho}{3}\right)2l \frac{\pi(4d)^2}{4}} = \frac{\frac{\rho l}{\pi d^2}}{\frac{2}{3}\rho l \frac{16\pi d^2}{16\pi d^2}} \quad [1 \text{ mark}]$$

- $\frac{\text{power dissipated in } Y}{\text{power dissipated in } X} = \frac{P_Y}{P_X} = 24$ [1 mark]

[Total: 3 marks]

The reason $P = I^2 R$ is used and not $P = \frac{V^2}{R}$ is because of the arranging of the resistors.

Since they are in **series**, they have the **same** current I but a different potential difference V . Using $P = I^2 R$ means that ' I ' cancels out in the ratio whilst V wouldn't.

1c

The ratio $\frac{\text{power dissipated in } Y}{\text{power dissipated in } X}$ of X and Y connected in series and connected in

parallel is:

List the known quantities:

- $\frac{P_Y}{P_{X_{series}}} = 24$ (from part b)

Determine $\frac{P_Y}{P_X}$ in parallel:

- $P = IV = \frac{V^2}{R}$ where $R = \frac{\rho l}{A}$
- $\frac{P_Y}{P_X} = \frac{\frac{V^2}{R_Y}}{\frac{V^2}{R_X}} = \frac{R_X}{R_Y} = \frac{\frac{\rho_X l_X}{A_X}}{\frac{\rho_Y l_Y}{A_Y}} \quad [1 \text{ mark}]$

Substitute the values from Table 1.1 into the equation and simplify:

- $\frac{P_Y}{P_X} = \frac{\frac{\left(\frac{\rho}{3}\right)2l}{\pi(4d)^2}}{\frac{\rho l}{\frac{\pi d^2}{4}}} = \frac{\frac{2}{3}\rho l}{\frac{\rho l}{\pi d^2}}$
- $\frac{P_Y}{P_{X_{parallel}}} = \frac{1}{24} \quad [1 \text{ mark}]$

Calculate the product of the ratios:

- $\frac{P_Y}{P_{X_{series}}} \times \frac{P_Y}{P_{X_{parallel}}} = 24 \times \frac{1}{24} = 1 \quad [1 \text{ mark}]$

[Total: 3 marks]

Both part (b) and (c) requires quite a bit of algebra here. You must be confident with dealing with fractions, as many of the harder questions are very algebraic. This style

of question is quite common in MCQs too.

2a

Calculate the current which passes through the carbon film in Fig. 1.1. for an applied voltage of 2.5 mV:

List the known quantities:

- Resistivity of carbon, $\rho = 4.0 \times 10^{-5} \Omega \text{ m}$
- Applied voltage, $V = 2.5 \text{ mV} = 2.5 \times 10^{-3} \text{ V}$

Interpret the correct dimensions from Fig. 1.1:

- Length which current passes through, $L = 3.0 \times 10^{-3} \text{ m}$

AND

- Cross-sectional area of conductor,
 $A = (3.0 \times 10^{-3}) \times (2.0 \times 10^{-6}) = 6.0 \times 10^{-9} \text{ m}^2$ [1 mark]

Use the resistivity equation to calculate the resistance R of the carbon film:

- $\rho = \frac{RA}{L}$
- $R = \frac{\rho L}{A} = \frac{(4.0 \times 10^{-5}) \times (3.0 \times 10^{-3})}{(6.0 \times 10^{-9})}$ [1 mark]
- $R = 20 \Omega$ [1 mark]

Use Ohm's Law to calculate the current flowing through the carbon film:

- $V = IR$
- $I = \frac{V}{R} = \frac{(2.5 \times 10^{-3})}{20}$ [1 mark]
- $I = 1.25 \times 10^{-4} \text{ A}$ **or** 0.13 mA [1 mark]

- Cross-sectional area of conductor,
 $A = (3.0 \times 10^{-3}) \times (2.0 \times 10^{-6}) = 6.0 \times 10^{-9} \text{ m}^2$ [1 mark]

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- $R = 20 \Omega$ [1 mark]

Use Ohm's Law to calculate the current flowing through the carbon film:

- $V = IR$
- $I = \frac{V}{R} = \frac{(2.5 \times 10^{-3})}{20}$ [1 mark]
- $I = 1.25 \times 10^{-4} \text{ A}$ **or** 0.13 mA [1 mark]

[Total: 5 marks]

2b

Show that the current is approximately a million times larger if it is directed through the carbon film as shown in Fig. 1.2:

List the known quantities:

- Resistivity of carbon, $\rho = 4.0 \times 10^{-5} \Omega \text{ m}$
- Applied voltage, $V = 2.5 \text{ mV} = 2.5 \times 10^{-3} \text{ V}$
- Current when directed horizontally, $I_A = 1.25 \times 10^{-4} \text{ A}$ (from part a)

Interpret the correct dimensions from Fig. 1.2:

- The length that the current passes through, $L = 2.0 \times 10^{-6} \text{ m}$

AND

- The cross-sectional area of conductor,
 $A = (8.0 \times 10^{-3}) \times (3.0 \times 10^{-3}) = 2.4 \times 10^{-5} \text{ m}^2$ [1 mark]

Use the resistivity equation to calculate the resistance R of the carbon film:

- $\rho = \frac{RA}{L}$
- $R = \frac{\rho L}{A} = \frac{(4.0 \times 10^{-5}) \times (2.0 \times 10^{-6})}{(2.4 \times 10^{-5})}$ [1 mark]
- $R = 3.3 \times 10^{-6} \Omega$ [1 mark]

Use Ohm's Law to calculate the current flowing through the carbon film, I_b :

- $V = IR$
- $I = \frac{V}{R} = \frac{(2.5 \times 10^{-3})}{(3.3 \times 10^{-6})}$
- $I_b = 756 \text{ A}$ [1 mark]

Calculate the ratio of currents:

- $\frac{I_b}{I_a} = \frac{756}{1.25 \times 10^{-4}} = 6.0 \times 10^6$ [1 mark]

State a conclusion:

- Therefore, the current in Fig. 1.2 is about a million times / 10^6 larger than the current in Fig. 1.1; [1 mark]

[Total: 6 marks]

2c

Discuss how the resistance of the carbon film changes as a result of the applied tensile force:

State how the resistance changes:

- The resistance decreases; [1 mark]

Describe the dimensional changes caused by the tensile force:

- The tensile force causes the length (of the carbon film through which current is flowing) to decrease **AND** the cross-sectional area to increase; [1 mark]

Explain how dimensional changes affects the resistance:

- (Resistance decreases because) resistance is proportional to length **AND** inversely proportional to the cross-sectional area; [1 mark]

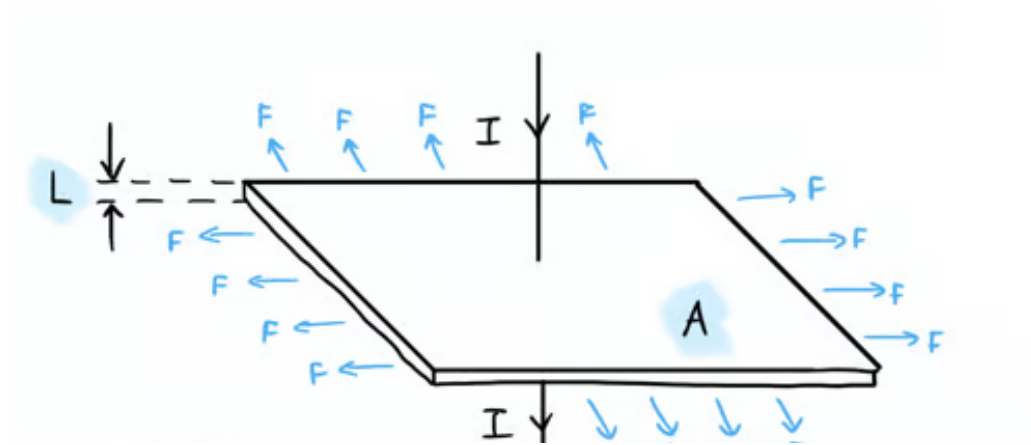
State any assumptions:

Any **one** of the following:

- The tensile force acts uniformly / is a constant magnitude; [1 mark]
- The tensile force acts in all directions; [1 mark]
- There are no tears / rips / deformities in the carbon film as it stretches; [1 mark]

[Total: 4 marks]

A **tensile** force is a **pulling force**, so the carbon film will be stretched by this force.



2c

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